Resonant Biofabrication and Quantum-Metrological Phenotyping: Topological Chiral Metamaterials, Terahertz Polarimetry, and Multimodal Embryonic Viability Assessment

1. Executive Summary and Technology Fusion Landscape

The assessment of mammalian gametes and preimplantation embryos within assisted reproductive technology (ART) remains constrained by a fundamental trade-off between diagnostic invasiveness and analytical resolution. Traditional clinical workflows depend almost entirely on subjective morphological grading or optical time-lapse incubation systems. Although automated optical time-lapse systems systematically track the kinematics of cellular division—such as cleavage intervals, cell cycle synchrony, and blastulation timing—they record only the visible macroscopic consequences of cellular metabolism and genomic integrity. By the time morphokinetic abnormalities become optically apparent, developmental arrest or cellular fragmentation is frequently underway. Conversely, direct biochemical and molecular assays, including preimplantation genetic testing for aneuploidies (PGT-A) via trophectoderm biopsy, necessitate mechanical breaches of the zona pellucida and cryopreservation cycles that introduce cellular trauma and clinical overhead.

A non-invasive, continuous diagnostic modality capable of mapping intracellular integrity in real time requires the fusion of several distinct physical regimes into a unified platform. The platform architecture analyzed herein integrates field-assisted additive micromanufacturing, biomimetic chiral nanocomposites, non-orientable topological resonators, near-field terahertz polarimetry, and differential quantum magnetometry.

The core manufacturing engine is governed by continuous Signed Distance Fields (SDF) and volumetric voxel representations that define structural geometry, internal fiber director orientation, and structural chirality as unified mathematical fields. This implicit formulation directly drives a micro-aperture extrusion printhead integrated with dynamic rotating electromagnetic fields ("roto-fields"). The system aligns high-performance aramid nanofibers (ANFs) and regenerated silk fibroin matrices in-situ, immediately locking the programmed anisotropy via coaxial photopolymerization. The target geometries manufactured by this process are topological Möbius strip metamaterials. Because a Möbius strip constitutes a non-orientable, single-sided manifold, an electromagnetic wave traversing its perimeter accumulates an intrinsic geometric Pancharatnam-Berry phase of \pi, breaking the degeneracy between clockwise and counter-clockwise modes and establishing fractional, half-integer topological resonances possessing strong circular chiroptical selectivity.

To interrogate living embryos without causing physiological harm or encountering the catastrophic absorption of liquid water, the platform deploys an opposed evanescent near-field

architecture. Operating in the sub-terahertz and terahertz bands (0.1\text{ to }2.5\text{ THz}), localized near-field decay tails couple into the cortical cytoskeleton, membrane hydration shells, and dividing blastomeres, extracting shifts in complex permittivity, quality factor (Q-factor), and terahertz circular dichroism (TCD) without penetrating bulk liquid media.

This electro-optical channel is supplemented by a room-temperature quantum magnetometry kernel adapted from mobile nitrogen-vacancy (NV) diamond platforms. Operating via optically detected magnetic resonance (ODMR), an integrated synthetic gradiometer executes real-time common-mode cancellation of ambient environmental baselines—specifically Earth-ionosphere Schumann resonances (7.83\text{ Hz} fundamental) and laboratory electromagnetic interference (EMI)—thereby isolating the faint pico-to-nanotesla bio-magnetic signatures associated with organellar ionic flux and mitotic division.

The resulting multi-stream data ingest feeds into a Multi-Modal Cross-Attention Temporal Transformer (MCAT-Net), which maps developmental trajectories against separate meiotic and mitotic spectral atlases to deliver predictive viability scores well in advance of optical cleavage anomalies.

2. Nanomaterials and Chiral Phononics: Kotov Aramid/Silk Composites and Terahertz Chiroptics

The physical transducer layer of this platform is anchored in the biomimetic nanomaterials developed by Nicholas Kotov and coworkers at the University of Michigan, particularly high-strength aramid nanofibers (ANFs) and fused silk fibroin nanocomposites. Natural silkworm (*Bombyx mori*) silk and spider dragline silk achieve high tensile strength and toughness through a hierarchical architecture composed of semi-crystalline anti-parallel \beta-sheet poly(alanine-glycine) nanocrystals embedded within an amorphous peptide matrix. Kotov's laboratory demonstrated that thermomechanical processing can fuse silk fibers into continuous, solid materials exhibiting flexural strengths up to approximately 510\text{ MPa} and exceptional tensile toughness (\approx 45\text{ MJ/m}^3). Simultaneously, Kotov's work on the controlled dissociation of macro-aramid plastics, such as poly(p-phenylene terephthalamide) (PPTA), demonstrated that synthetic aramid nanofibers can replicate the extensive sacrificial hydrogen-bonding networks of biological silk while providing superior chemical resistance, thermal stability, and mechanical stiffness.

In high-frequency electromagnetic and terahertz sensing, silk fibroin functions not merely as an inert structural carrier, but as an active, tunable dielectric transducer. Within the 0.1\text{ to }3.0\text{ THz} spectral window, dried regenerated silk fibroin exhibits an optical refractive index ranging from n \approx 1.91 to 2.1, combined with low dielectric loss compared to synthetic hydrated polymers. When deposited onto high-frequency metamaterials, such as metallic split-ring resonators, silk fibroin layers induce pronounced, thickness-dependent red-shifts in fundamental inductive-capacitive (LC) resonant frequencies, directly reflecting changes in local dielectric permittivity and atmospheric hydration dynamics. The peptide backbone of silk fibroin undergoes reversible and irreversible conformational transitions between disordered random coils and organized crystalline \beta-sheets when subjected to external electric fields, shear stress, and localized dehydration. This structural malleability allows the silk matrix to match electromagnetic impedances between high-index semiconductor or metallic transmission lines and aqueous biological environments.

Beyond bulk dielectric behavior, the crucial physical phenomenon governing high-frequency biological selectivity is the excitation of **chiral phonons** and **chiroplasmonic resonance** within

the terahertz band. Chiral phonons represent collective, mirror-symmetry-broken lattice vibrations possessing non-zero angular momentum. In chiral biocrystals, including amino acids, oligopeptides, and cytoskeletal polymers, low-energy lattice vibrations between 0.5 and 3.5\text{ THz} correspond to the coherent twisting and rotational motions of molecular segments involving dozens of atoms, with eigenfrequencies highly sensitive to molecular stereochemistry and local hydrogen-bonding architecture. These chiral phonons couple selectively to circularly polarized terahertz radiation, generating distinct terahertz circular dichroism (TCD) bands and optical rotation dispersion (TORD) spectra.
Kotov and collaborators demonstrated this principle using Arrays of Chiral Microneedles (ARCHIMs) fabricated via sequential glancing-angle gold deposition. When functionalized with chiral biocrystalline overlays (such as L- and D-cystine), the ARCHIM platforms exhibited strong chiral plasmonic modes that resonated directly with the chiral phonons of the biocrystals. This chiral-plasmon-chiral-phonon coupling induced handedness-dependent frequency shifts and polarization rotations exceeding 5^\circ, which are quantitatively described by temporal coupled-mode theory. In the ScanIQ platform, incorporating synthetic aramid/silk nanocomposites embedded with chiral crystalline domains allows the 3D-printed resonator to act as an active chiroptical filter. The internal molecular handedness of the composite hybridizes with the higher-order evanescent fields of the resonator, amplifying minute chiroptical changes in an adjacent embryo's macromolecular framework.

3. Field-Assisted Additive Manufacturing of Topological Resonators

Mathematical Formulation: Signed Distance Fields and Material Tensor Fields

Conventional computer-aided design (CAD) systems represent mechanical objects using boundary representations (B-rep) and surface-tessellated polygon meshes. These boundary-centric models are incapable of representing continuous, functionally graded material variations, localized fiber orientation fields, or sub-millimeter crystalline chirality.
To overcome these structural limitations, the fabrication pipeline utilizes continuous Signed Distance Fields (SDF) unified with volumetric material tensor fields. The spatial boundary of the target geometry \Omega \subset \mathbb{R}^3 is defined implicitly by a scalar distance function f(\vec{x}):
f(\vec{x}) = \text{dist}(\vec{x}, \partial\Omega) \cdot \begin{cases} -1 & \text{if } \vec{x} \in \Omega \\ +1 & \text{if } \vec{x} \notin \Omega \end{cases}
Co-registered across the spatial domain is a continuous material tensor field \mathbf{M}(\vec{x}) that dictates local physical composition at every point:
\mathbf{M}(\vec{x}) = \left\{ \phi_f(\vec{x}), \, \vec{v}_{\text{fiber}}(\vec{x}), \, \chi(\vec{x}), \, \epsilon_r(\vec{x}), \, \sigma_e(\vec{x}) \right\}
where \phi_f(\vec{x}) \in [0, 1] represents the local volume fraction of reinforcing silk or aramid nanofibers; \vec{v}_{\text{fiber}}(\vec{x}) \in \mathbb{S}^2 is a unit vector specifying the spatial orientation of the fiber director; \chi(\vec{x}) \in [-1, +1] quantifies the local chiral enantiomeric excess or structural handedness; \epsilon_r(\vec{x}) represents the complex dielectric permittivity tensor; and \sigma_e(\vec{x}) defines localized electrical conductivity. By utilizing implicit, gradient-informed slicing algorithms, fabrication toolpaths and orientation vectors are

computed directly from these field equations, eliminating intermediate polygon conversions and facilitating sub-micron control over internal structural gradients.

Aperture Printhead and Dynamic Electromagnetic "Roto-Fields"

Translating the material tensor field $\mathbf{M}(\vec{x})$ into physical hardware requires an additive manufacturing printhead capable of steering anisotropic fibrils during extrusion. The liquid feedstock consists of a photocrosslinkable prepolymer resin—such as methacrylated silk fibroin (Sil-MA) blended with functionalized aramid nanofibers and water-soluble photoinitiators—carrying suspended, high-aspect-ratio anisotropic fibrils or magnetically functionalized nanoparticles.

The deposition hardware features a vertical micro-aperture nozzle ($50\text{ to }250\text{ }\mu\text{m}$ inner diameter) surrounded by an active electromagnetic orientation collar. This collar consists of a multi-pole micro-solenoid stator array and electrostatic deflection plates configured to generate high-speed rotating vector fields—termed **roto-fields**—directly within the vertical print zone. The dynamic field equations are expressed as:

$$\vec{B}_{\text{roto}}(t) = B_0 \left[\cos(\omega_r t + \psi) \hat{i} + \sin(\omega_r t + \psi) \hat{j} \right] + B_z \hat{k} \vec{E}_{\text{roto}}(t) = E_0 \left[\cos(\omega_e t + \delta) \hat{i} + \sin(\omega_e t + \delta) \hat{j} \right] + E_z \hat{k}$$

As the composite fluid traverses the nozzle aperture, the external rotating fields generate hydrodynamic and magnetic torques ($\vec{\tau} = \vec{p} \times \vec{E} + \vec{m} \times \vec{B}$) that align the anisotropic fibrils along the field direction. By dynamically modulating the rotational frequencies (ω_r, ω_e), phase offsets (ψ, δ), and vertical printhead translation rate v_z, the system programs the three-dimensional fiber director $\vec{v}_{\text{fiber}}(\vec{x})$ within each deposited voxel. Coaxial UV/blue LED arrays (365 nm / 405 nm) positioned around the nozzle tip trigger rapid in-situ photopolymerization, crosslinking the polymer matrix within milliseconds to lock the programmed molecular alignment and structural chirality into place before Brownian motion can induce relaxation.

Topological Möbius Strip Resonator Physics

The geometric target produced by this field-assisted additive process is a non-orientable topological structure: the **Möbius strip resonator**. A Möbius strip is defined by taking a ribbon of physical circumference L, width w, and thickness t, twisting one end through an odd integer multiple of 180° (π radians), and seamlessly joining the ends. The resulting geometry possesses only a single continuous boundary curve and a single non-orientable surface.

In a standard, untwisted closed ring resonator of circumference L, constructive interference occurs when the round-trip dynamic phase accumulation equals an integer multiple of 2π, establishing conventional whispering-gallery or Fabry-Pérot resonant frequencies:

$$\beta L = 2m\pi \quad \implies \quad f_m = \frac{m \cdot c}{n_{\text{eff}} L}, \quad m \in \mathbb{Z}^+$$

In a Möbius strip resonator, the non-orientable boundary conditions force the transverse polarization vector to undergo a continuous 180° spatial inversion over a single physical circuit L. Consequently, an electromagnetic wave must execute *two complete orbital revolutions* (2L) to return to its initial vector state, accumulating an intrinsic geometric **Pancharatnam-Berry phase** $\gamma = \pi$. The resonant condition is thereby modified according to:

$$\beta (2L) + \gamma = 2\pi N \quad \implies \quad \beta L = \left(m + \frac{1}{2} \right) \pi$$

This non-trivial topological boundary condition shifts the allowable longitudinal eigenmodes to

fractional, half-integer modes. Furthermore, because a surface normal vector cannot be globally defined across a non-orientable manifold, the physical symmetry between clockwise (CW) and counter-clockwise (CCW) propagation is broken. Circularly polarized modes experience asymmetric phase accumulation depending on whether their electromagnetic handedness aligns with or opposes the physical structural twist of the strip. As demonstrated by Chen et al. (2023), this causes the Möbius resonator to resonate selectively with specific chiral circular polarization states, transforming the physical strip into a high-Q, chirality-selective topological filter.

The additive manufacturing platform aligns the internal aramid/silk nanofiber director $\vec{v}_{\text{fiber}}(\vec{x})$ along the helical centerline of this Möbius manifold. This spatial structuring eliminates internal scattering boundaries, minimizes dielectric radiation loss, and elevates the intrinsic quality factor ($Q = \omega_0 \mathcal{E}_{\text{stored}} / P_{\text{diss}}$), producing an electromagnetic cavity highly sensitive to external dielectric perturbations.

Metamaterial / Resonator Geometry	Topological Class & Berry Phase	Longitudinal Mode Spectrum Spacing	Chiral & Polarization Sensitivity	Substrate & Mechanical Characteristics
Planar Split-Ring Resonator (SRR)	Trivial ($\gamma = 0$); Dual-sided orientable	Integer Fabry-Pérot harmonics: $f_m \propto m$	Linear polarization along capacitive gap axis	Lithographic Au on Silicon or planar silk film
Standard Ring Resonator (RR)	Trivial ($\gamma = 0$); Dual-sided orientable	Degenerate CW / CCW modes: $f_m \propto m$	Isotropic; polarization-degenerate	Silica optical fiber or extruded polymer ring
Topological Möbius Strip (ScanIQ)	Non-trivial ($\gamma = \pi$); Single-sided non-orientable	Fractional half-integer modes: $f_m \propto (m + 1/2)$	Chirality-selective; lifts CW/CCW degeneracy	Roto-field aligned fused silk/ANF composite (510 MPa)
Chiral Microneedle Array (ARCHIM)	Trivial topology; Broken mirror symmetry	Hybridized chiral plasmon-phonon bands	Strong optical rotation dispersion ($>5^\circ$)	Micro-machined Si / polymer with angled Au caps

4. Terahertz Pulse Modulation and Evanescent Near-Field Polarimetry

Deploying terahertz radiation ($0.1\text{ to }10\text{ THz}$) for developmental biological sensing introduces an immediate physical challenge: liquid water exhibits an exceptionally high absorption coefficient across this spectrum ($\alpha_{\text{water}} \approx 150\text{ to }300\text{ cm}^{-1}$ at 1 THz). Direct, bulk transmission of a terahertz beam through standard physiological culture droplets ($>500\text{ }\mu\text{m}$ thickness) results in total signal extinction, masking cellular spectra.

To circumvent this limitation, the ScanIQ platform replaces bulk transmission with an **opposed evanescent near-field coupling configuration**. Two 3D-printed topological Möbius strip resonators are mounted in mirror opposition directly above and below a low-loss cyclo-olefin polymer (COP) or thin fused-silk microfluidic sensing cassette. The fluid channel height is constrained to between 100 and $170\text{ }\mu\text{m}$, matching the physical dimensions of mammalian oocytes and blastocysts while minimizing media volume. Interrogation relies strictly on the evanescent field decay tails extending outward from the resonator boundaries into the

microchannel:
$$\vec{E}_{\text{evanescent}}(z) = \vec{E}_0 e^{-\kappa z}, \quad \kappa = \sqrt{\beta^2 - k_0^2 \epsilon_{\text{channel}}}$$
Because the penetration depth of this evanescent field is confined to several micrometers ($z < 10\text{ }\mu\text{m}$), the electromagnetic field selectively probes the outer zona pellucida, plasma membrane, and cortical cytoskeleton of the specimen without penetrating into deep, lossy liquid media.
The signal generation and detection chain is structured as follows:

- **Pulse Generation and Encoding**: An ultrafast femtosecond fiber laser ($\lambda = 1560\text{ nm}$, pulse width $\tau < 50\text{ fs}$) drives an array of photoconductive antennas (PCAs) to emit broadband terahertz pulses spanning $0.2\text{ to }2.5\text{ THz}$. The terahertz pulse train is modulated via high-speed liquid crystal phase retarders, alternating between left-handed circularly polarized (LCP, σ_+) and right-handed circularly polarized (RCP, σ_-) states.
- **Evanescent Cavity Coupling**: The polarization-encoded pulse is coupled via a dielectric horn into the upper Möbius strip resonator, generating an intense, localized near-field within the microfluidic incubation well.
- **Cavity Perturbation**: The presence of the developing embryo alters the local boundary conditions, shifting the complex eigenfrequency ($\tilde{\omega} = \omega_0 + \Delta\omega + i \Delta\Gamma$) according to cavity perturbation theory:

$$\frac{\Delta\omega}{\omega_0} + i \frac{\Delta(1/2Q)}{1} \approx -\frac{\int_{V_s} \left(\Delta\epsilon |\vec{E}_0|^2 + \Delta\mu |\vec{H}_0|^2 \right) dV}{2 \int_{V_c} \left(\epsilon_r |\vec{E}_0|^2 + \mu_r |\vec{H}_0|^2 \right) dV}$$

where V_s is the biological specimen volume, V_c is the cavity volume, $\Delta\epsilon = (\epsilon_{\text{embryo}} - \epsilon_{\text{media}})$ is the differential complex permittivity perturbation, and ΔQ is the change in quality factor resulting from dielectric absorption.

- **Polarimetric Detection**: The perturbed pulse emerging from the opposed lower resonator is captured via electro-optic sampling (EOS) utilizing a thin ZnTe crystal and balanced photodiode bridge. Orthogonal electric field components ($E_x(t)$, $E_y(t)$) are simultaneously digitized, yielding the full time-resolved Stokes vector $\vec{S}(t) = [S_0, S_1, S_2, S_3]^T$:

$$S_0 = |E_[span_150](start_span)[span_150](end_span)x|^2 + |E_y|^2, \quad S_1 = |E_x|^2 - |E_y|^2 \quad S_2 = 2 \text{Re}(E_x E_y^*), \quad S_3 = 2 \text{Im}(E_x E_y^*)$$

The differential parameter $\Delta S_3(\omega) = S_3^{\text{LCP}}(\omega) - S_3^{\text{RCP}}(\omega)$ extracts the pure terahertz circular dichroism (TCD) and optical rotation induced by the biological specimen, separating surface structural changes from environmental drift.

5. Quantum Magnetometry Integration and Environmental Baseline Denoising

Nitrogen-Vacancy Diamond Vector Magnetometry

To provide an orthogonal physical measurement that complements dielectric and polarimetric sensing, the platform integrates a quantum magnetometry subsystem derived from the mobile nitrogen-vacancy (NV) diamond sensing architecture. The negatively charged NV center

(\text{NV}^-) is a point defect within the diamond carbon lattice comprising a substitutional nitrogen atom adjacent to a lattice vacancy, possessing a spin triplet ground state (S = 1). The ground-state spin Hamiltonian is given by:
\hat{H}_{\text{NV}} = h D \hat{S}_z^2 + g_e \mu_B \vec{B} \cdot \hat{\vec{S}} + h E \left(\hat{S}_x^2 - \hat{S}_y^2 \right)
where D \approx 2.87\text{ GHz} represents the temperature-dependent zero-field splitting parameter separating the |m_s = 0\rangle and |m_s = \pm 1\rangle spin sublevels; g_e \approx 2.0028 is the Landé g-factor; \mu_B is the Bohr magneton; \vec{B} is the local magnetic field vector; and E is the non-axial strain parameter.
Optically Detected Magnetic Resonance (ODMR) is executed by optically pumping the NV diamond layer with a continuous-wave green laser (\lambda = 532\text{ nm}), which polarizes the electron spin ensemble into the bright |m_s = 0\rangle ground state. Sweeping a transverse microwave field across the 2.87\text{ GHz} resonance drives transitions into the |m_s = \pm 1\rangle states, which decay via a non-radiative intersystem crossing to a metastable singlet state, producing a pronounced reduction in emitted red fluorescence (637\text{ to }800\text{ nm}). When an external magnetic field \vec{B} is applied, the Zeeman effect lifts the degeneracy between the |m_s = +1\rangle and |m_s = -1\rangle sublevels, splitting the ODMR resonance dips by:
\Delta f_{\pm} = 2 \gamma_e B_z, \quad \gamma_e = \frac{g_e \mu_B}{h} \approx 28\text{ Hz/nT}
By measuring this Zeeman splitting along the four crystallographic tetrahedral axes (\langle 111 \rangle) of the diamond lattice, the system reconstructs full vector magnetic fields [\Delta B_x, \Delta B_y, \Delta B_z] with sub-nanotesla sensitivity at physiological room temperatures.

Gradiometric Baseline Denoising via Schumann Resonances

Detecting biological magnetic fields (typically pico- to nano-tesla fluctuations generated by trans-membrane ionic currents, metabolic redox reactions, and mitotic spindle movement) requires overcoming severe environmental magnetic noise. Ambient backgrounds are dominated by the Earth's static geomagnetic field (30\text{ to }60\text{ }\mu\text{T}), power-grid harmonics (50/60\text{ Hz}), mechanical laboratory vibrations, and planetary electromagnetic standing waves known as **Schumann resonances**.
Schumann resonances are continuous, extremely low frequency (ELF) transverse magnetic waves excited by global lightning activity (\approx 40\text{ to }50\text{ events/sec}) within the spherical cavity formed by the Earth's surface and the lower ionosphere. The fundamental mode occurs at:
f_n \approx \frac{c}{2\pi R_{\text{earth}}} \sqrt{n(n+1)}, \quad f_1 \approx 7.83\text{ Hz}, \quad f_2 \approx 14.1\text{ Hz}, \quad f_3 \approx 20.3\text{ Hz}
In sensitive metrology, Schumann resonances generate ambient background magnetic power densities of 0.5\text{ to }2.0\text{ pT}/\sqrt{\text{Hz}} across the 5\text{ to }40\text{ Hz} band, punctuated by ELF transient bursts.
Rather than attempting to eliminate these global resonances with bulky, high-permeability magnetic shielding (which disrupts incubator thermal and gas dynamics), the system deploys a dual-node **synthetic reference gradiometer**:

- **Sensing Node**: An integrated NV diamond layer positioned directly beneath the microfluidic embryo trap, capturing the combined bio-magnetic field and ambient background: \vec{B}_{\text{sens}}(t) = \vec{B}_{\text{bio}}(t) + \vec{B}_{\text{ambient}}(t).
- **Reference Node**: An identical NV diamond node positioned within the same incubator

housing but laterally offset by an isolation baseline ($d \approx 30\text{ mm}$), where biological dipole coupling is zero due to the steep spatial roll-off ($\propto 1/r^3$), recording pure $\vec{B}_{\text{ambient}}(t)$.

Raw time-tagged vector ODMR traces from both nodes are processed by an on-board FPGA executing an adaptive finite impulse response (FIR) Wiener filter and multi-resolution wavelet shrinkage. Treating the coherent planetary Schumann resonance modes (7.83 Hz, 14.1 Hz, etc.) as common-mode reference baselines, the adaptive filter subtracts ambient correlated noise:

$$\vec{B}_{\text{bio_filtered}}(t) = \vec{B}_{\text{sens}}(t) - \sum_{k=0}^{P} \mathbf{W}_k \vec{B}_{\text{ref}}(t-k)$$

This differential architecture suppresses geomagnetic drift and power-line interference by $>85\text{ dB}$, enabling clear extraction of cellular and organellar bio-magnetic transients.

6. Embryonic Resonance, Dielectric Mapping, and Developmental Phenotyping

Empirical Foundations in Electrical Impedance Spectroscopy

The high-frequency dielectric and quantum phenotyping channels are grounded directly in published experimental data from Differential Electrical Impedance Spectroscopy (EIS) performed on mammalian embryos and oocytes.

A primary quantitative anchor is provided by the research of Uzzaman, Rahman, and Ohta (IEEE EMBC 2024), who investigated bovine embryo development within a microfluidic impedance trap across frequencies from 1 kHz to 5 MHz. Their measurements demonstrated a statistically significant decrease in the real component of differential impedance ($\Delta \text{[span_37](start_span)[span_37](end_span)Re}(Z)$) localized at a distinct relaxation frequency of **1.49 MHz**. This 1.49 MHz decrement correlated directly with increasing cellular mass: the normalized cell mass expanded by 11% at 75 minutes, 38% at 150 minutes, and 91% at 225 minutes, tracking the differential electrical current shunted through multiplying blastomere membranes.

These findings are corroborated by Alalul, Liu, Böl, Al-Halhouli, and Dietzel (2026), who conducted single-cell dielectric and mechanical phenotyping of porcine oocytes up to 30 MHz. Their research showed that gamete physiological integrity, cytoplasmic degradation, and cryopreservation injury cause massive, measurable deviations in high-frequency capacitive reactance, directly reflecting changes in membrane intactness and cortical cytoskeleton elasticity.

Multi-Scale Bio-Electromagnetic Dispersion Physics

The dielectric response of developing cellular systems spans several well-defined dispersion regimes characterized by the Cole-Cole relaxation model:

$$Z(\omega) = R_\infty + \frac{R_0 - R_\infty}{1 + (j\omega \tau)^\alpha}$$

where R_0 is the low-frequency resistance, R_∞ is the high-frequency limit, τ is the characteristic relaxation time, and α parameterizes distribution broadening.

- **α-Dispersion ($10\text{ Hz} - 10\text{ kHz}$)**: Governed by the tangential diffusion and relaxation of ionic counter-charges within the electrical double layer (Debye screening

layer) surrounding the outer plasma membrane and zona pellucida.

- **\beta-Dispersion (100\text{ kHz} - 50\text{ MHz})**: Governed by Maxwell-Wagner interfacial polarization, where intact lipid bilayers act as thin insulating capacitors (C_{\text{mem}} \approx 1.0\text{ }\mu\text{F/cm}^2) separating conductive extracellular and intracellular electrolytes. At frequencies below 1\text{ MHz}, electrical current is largely diverted around cells. Between 1\text{ MHz[span_192](start_span)[span_192](end_span)[span_195](start_span)[span_195](end_span)} and 10\text{ MHz}—encompassing the Uzzaman 1.49\text{ MHz} marker—current capacitively crosses the plasma membrane, directly probing intra-cytoplasmic conductivity (\sigma_{\text{cyto}}), blastomere cleavage planes, and intercellular tight junctions.
- **\gamma-Dispersion and Sub-THz Band (0.1\text{ GHz} - 2\text{ THz})**: Driven by the rotational relaxation of free and biologically bound water dipoles. As cleavage progresses to morula compaction and blastocoele fluid accumulation, the ratio of bound hydration water (structured around proteins and cytoskeletal polymers) to free bulk water shifts dramatically, altering high-frequency dielectric loss and loaded cavity Q-factors.
- **Chiral Phonon Window (1.0 - 3.5\text{ THz})**: Probes collective twisting vibrations within macromolecular assemblies (such as mitochondrial inner-membrane complexes and mitotic spindles), selectively modulating the polarization handedness of circularly polarized terahertz pulses.

Developmental Stage / Biological State	Morphological & Cytoskeletal Features	Dominant Dielectric & Impedance Signatures	Topological Resonator & Q-Factor Response	Gradiometric NV Magnetic Observable
Immature Oocyte (GV / MI Stage)	Large central germinal vesicle; dense actin cortex; absent polar body.	Low membrane capacitance (C_m \approx 0.8\text{ }\mu\text{F/cm}^2); high low-frequency capacitive reactance.	Narrow baseline resonance dip; minimal dielectric dissipation (Q \approx Q_0).	Quiescent; flat magnetic spectrum matching filtered geomagnetic background.
Mature Oocyte (Metaphase II)	Extruded first polar body; meiotic spindle assembled at cortex.	Elevated membrane capacitance (C_m \approx 1.[span_187](start_span)[span_187](end_span)1\text{ }\mu\text{F/cm}^2); shifted \beta-dispersion inflection.	Slight handedness-dependent phase rotation from aligned meiotic spindle polymers.	Pulsatile localized ionic currents associated with calcium priming.
Zygote (Pronuclear Stage, Day 1)	Two pronuclei (2PN) visible; cortical granule exocytosis complete.	Intermediate real impedance; transient \alpha-dispersion fluctuations during	Moderate Q-factor damping; initial dielectric loading shift.	Single high-amplitude calcium-induced magnetic pulse transient (<5\text{

Developmental Stage / Biological State	Morphological & Cytoskeletal Features	Dominant Dielectric & Impedance Signatures	Topological Resonator & Q-Factor Response	Gradiometric NV Magnetic Observable
		pronuclear migration.		nT}).
Early Cleavage (2- to 8-Cell, Days 2-3)	Distinct spherical blastomeres; active mitotic spindle assembly and cytokinesis.	Stepwise decrement in \Delta \text{Re}(Z) at 1.49\text{ MHz} proportional to cell mass proliferation.	Cyclic Q-factor oscillations matching mitotic cycles; step shifts in resonance frequency \Delta f_r.	Periodic magnetic micro-bursts correlating with cytokinesis and microtubule sliding.
Morula (Compaction, Day 4)	Blastomeres flatten and maximize intercellular contact; tight junctions form.	Rapid reduction in extracellular current pathways; marked rise in low-frequency resistance (R_0).	Maximum dielectric perturbation per cell volume; broad resonance notch width.	Steady-state metabolic current rise due to active \text{Na}^+/\text{K}^+-ATPase pump assembly.
Expanded Blastocyst (Days 5-6)	Distinct ICM and trophectoderm; large fluid-filled blastocoele cavity.	Real impedance reaches minimum; major \gamma-dispersion shift from blastocoele water volume.	Maximum red-shift in topological mode (\Delta f_r > 50\text{ GHz}); strong Q-damping.	Sustained vector dipole field from polarized transepithelial ionic transport.
Developmentally Arrested / Aneuploid Embryo	Blastomere fragmentation, cytoplasmic blebbing, cell cycle arrest.	Impedance fails to follow 1.49\text{ MHz} curve; premature membrane depolarization.	Irregular, non-monotonic Q-factor fluctuations; absence of cyclic fractional mode shifts.	Absence of coherent periodic magnetic transients; high-entropy magnetic noise floor.

7. Artificial Intelligence Discriminator and Trajectory Modeling

Multi-Modal Data Ingestion and Synchronization

The ScanIQ diagnostic engine rejects the scientifically unsupported notion of a single "magic frequency" capable of stimulating or assessing embryo development. Instead, it conceptualizes embryogenesis as a continuous, deterministic trajectory traversing a high-dimensional electromechanical and morphokinetic phase space.
The acquisition architecture continuously ingests synchronized time-series data from five distinct physical modalities:

- **Electrical Impedance Spectra**: $Z(f, t) \in \mathbb{C}^{M}$, sampled across $M = 128$ logarithmic points from 1 kHz to 50 MHz, explicitly tracking the 1.49 MHz

developmental relaxation marker.

- **Topological Metamaterial Parameters**: Complex resonant frequency shifts \Delta f_r(t), loaded quality factors Q(t), resonance bandwidths \Delta\omega(t), and return losses S_{11}(f, t).
- **Terahertz Polarimetric Vectors**: Time-resolved Stokes parameters [S_0, S_1, S_2, S_3](t) and differential circular dichroism \Delta S_3(\omega, t) across the 0.2\text{ to }2.5\text{ THz} band.
- **Gradiometrically Filtered Magnetic Vectors**: Vector components \vec{B}_{\text{bio}}(t) = [B_x, B_y, B_z[span_132](start_span)[span_132](end_span)](t) acquired via NV diamond ODMR at a sampling rate of 250\text{ Hz}.
- **Optical Morphokinetics**: Brightfield time-lapse optical images \mathbf{I}(x, y, t), capturing spatial blastomere geometry, cleavage intervals (t_2, t_3, t_4, t_8), compaction timing (t_{\text{comp}}), and blastulation onset (t_{\text{sb}},[span_155](start_span)[span_155](end_span)[span_159](start_span)[span_159](end_span) t_b).
- **Environmental Telemetry Control**: Ambient temperature T(t), gas concentrations (\text{CO}_2, \text{O}_2), media pH, and common-mode EMI, serving as invariant null baselines to ensure the model does not misinterpret incubator fluctuations as embryonic state changes.

Deep Neural Network Architecture: MCAT-Net

The core computational model is structured as a Multi-Modal Cross-Attention Temporal Transformer (MCAT-Net).

The feature extraction stage utilizes modality-specific encoders. Raw impedance spectra and terahertz polarimetric vectors are processed through 1D convolutional residual networks (ResNet-1D) to isolate localized dielectric relaxation peaks and chiroptical dispersion inflections. Time-lapse optical frames are processed through a Spatial Vision Transformer (ViT) incorporating cross-membrane blastomere blending to segment cell boundaries and track volume changes. The vector magnetic field stream is processed through a continuous wavelet transform to isolate transient cellular current events. Concurrently, environmental telemetry vectors are projected into a latent subspace governed by an adversarial orthogonality loss, ensuring that internal feature representations remain mathematically invariant to external incubator drift.

The multimodal representations are merged within a cross-modal attention fusion layer. Bidirectional cross-attention mechanisms allow dielectric and magnetic representations to attend directly to visual cleavage milestones. This enables the model to correlate an impending cleavage furrow visible at time t with the preceding 1.49\text{ MHz} real impedance inflection and localized micro-magnetic transient.

The resulting sequence of fused tokens passes into an 8-layer temporal transformer backbone operating with a causal self-attention mask. The network tracks developmental trajectories across the entire 120\text{-hour} culture period, maintaining sensitivity to rate-of-change dynamics rather than static thresholds.

Separation of Meiotic and Mitotic Spectral Atlases

To prevent biological confounding, the training regimen strictly segregates the cellular division data into two independent spectral atlases:

- **Gamete Meiotic Atlas**: Calibrated exclusively on unfertilized oocytes undergoing *in vitro* maturation (IVM) through Germinal Vesicle Breakdown (GVBD), Metaphase I, and Metaphase II. This atlas isolates the dielectric and polarimetric signatures of reductive meiotic division, polar body extrusion, and cortical actin remodeling.
- **Embryonic Mitotic Cleavage Atlas**: Calibrated exclusively on post-fertilization development from zygote to expanded blastocyst. Because mitotic divisions involve rapid cleavage without cellular growth phases (G_1/G_2), they generate dramatic increases in total membrane surface area per unit volume, which produce characteristic stepwise decrements in the 1.49\text{ MHz} differential impedance channel.

Multi-Task Output Head

The transformer backbone terminates in three synchronized output heads:

1. **Developmental Stage Classifier**: Evaluates the instantaneous developmental milestone across a 13-stage progression model (from 1-cell zygote to hatched blastocyst).
2. **Trajectory Deviation Metric (\mathcal{D}_{\text{traj}}(t[span_134](start_span)[span_134](end_span)))**: Computes the Mahalanobis distance between the observed multidimensional trajectory and the normative manifold of healthy development. If an embryo exhibits abnormal ionic leakage or membrane depolarization, \mathcal{D}_{\text{traj}}(t) diverges significantly up to 18\text{ hours} before optical fragmentation or arrest becomes detectable by an embryologist.
3. **Viability & Competence Predictor**: Outputs a continuous probability score \mathcal{V}_{\text{comp}} \in [0.0, 1.0] quantifying euploidy likelihood and blastocyst developmental potential, targeting an area under the receiver operating characteristic curve (\text{AUC}) of >0.92 when validated against blinded clinical outcomes.

8. Pitch Deck Presentation: ScanIQ Bio-Resonance Platform

Slide 1: Cover Slide

- **Slide Title**: ScanIQ Bio-Resonance Platform
- **Subtitle**: Non-Invasive Developmental Metrology Engineered from Computationally Manufactured Resonators, Multimodal Physical Sensing, and Temporal AI
- **Design Hierarchy & Visual Directives**: Premium deep obsidian/navy background (#0A0E17) with crisp white typography and electric-teal geometric accents. Central visual displays a high-resolution photorealistic 3D rendering of an opposed dual-Möbius resonator well in translucent cyclo-olefin polymer, overlaid with faint cyan terahertz field lines and deep crimson optical fluorescence from the quantum diamond substrate.
- **Slide Content & Context**: ScanIQ LLC Research & Commercialization Briefing. Confidential concept document establishing a new physical metrology paradigm for assisted reproduction.

Slide 2: The Unmet Clinical Challenge in Assisted Reproduction

- **Slide Title**: The Diagnostic Blind Spot in Embryo Selection
- **Core Message**: Current embryo selection relies on subjective visual grading or invasive biopsies, leaving a critical gap for continuous, passive physical metrology.
- **Visual Layout**: Three-column comparative structural card layout over a dark slate background, contrasting current clinical standards against the ScanIQ continuous physical sensing approach.

Diagnostic Modality	Operational Mechanism	Clinical Limitations	ScanIQ Physical Layer
Optical Time-Lapse (Image-Led)	Evaluates appearance and cleavage kinematics via microscopy.	Subjective; accuracy plateaus at 60–65%; detects failure only after physical arrest.	Continuous passive dielectric and quantum sensing detects metabolic failure hours earlier.
Preimplantation Genetic Testing (PGT-A)	Invasive trophectoderm biopsy followed by NGS sequencing.	Introduces cellular micro-trauma; requires mandatory cryopreservation; adds $3,000–$5,000 cost.	Fully non-invasive; maintains intact zona pellucida; zero biopsy or freezing required.
Metabolic Spent Media Assays	Episodic measurement of spent culture droplet metabolites.	High dilution error; non-continuous; sensitive to ambient incubator evaporation.	Integrated microfluidic cassette provides continuous real-time impedance and Q-factor tracking.

Slide 3: Scientific and Empirical Foundation

- **Slide Title**: Validated Empirical Anchors
- **Core Message**: ScanIQ builds directly upon published, peer-reviewed breakthroughs in microfluidic impedance, nanomaterials, and terahertz spectroscopy.
- **Visual Layout**: Horizontal metric banner displaying four core empirical anchors, flanked by detailed scientific narrative cards.
- **Key Metric Highlights**:
  - **1.49 MHz**: Bovine Embryo Differential EIS Relaxation Marker (Uzzaman et al., IEEE EMBC 2024).
  - **13 Stages**: Developmental Milestones Mapped via Dynamic Impedance Trajectories.
  - **0.92 AUC**: Predictive Accuracy Target for Developmental Competence via Multimodal Fusion.
  - **510 MPa**: Flexural Strength of Michigan Kotov Fused Silk Nanocomposite Resonators.
- **Supporting Narrative**: Bovine embryo studies demonstrate that normalized cell mass expansion (11\% at 75\text{ min} \to 91\% at 225\text{ min}) directly correlates with a decrease in differential real impedance at 1.49\text{ MHz}. Concurrently, University of Michigan research proves that fused silk composites deliver the structural stiffness and low dielectric loss needed for stable high-Q metamaterials.

Slide 4: The Five-Stage Closed-Loop Platform Architecture

- **Slide Title**: Closed-Loop Computational Biofabrication & Sensing

- **Core Message**: A unified computational engine that links mathematical geometry to additive manufacturing, physical metrology, AI phenotyping, and iterative redesign.
- **Visual Layout**: Circular process flow diagram linking five primary stages with illuminated directional vectors and clean typography.
- **Process Stages**:
    1. *SDF & Material Tensor Field*: Mathematical definition of boundary geometry, fiber director vector, and chirality without polygon meshes.
    2. *Field-Oriented Additive Printing*: Micro-aperture deposition with dynamic electromagnetic roto-fields and in-situ UV cure.
    3. *Differential Multi-Physical Metrology*: Opposed resonators capture Z(f), \Delta f_r, \Delta Q, and vector magnetic field components.
    4. *Temporal Multimodal AI*: Temporal cross-attention transformer models developmental trajectories, stage classification, and viability.
    5. *Measured Redesign Loop*: Observed electromagnetic transfer functions update the continuous material field for subsequent print optimization.

Slide 5: Topological Resonator Physics: The Möbius Advantage

- **Slide Title**: Exploiting Non-Orientable Metamaterial Physics
- **Core Message**: Single-sided Möbius strip topology introduces an intrinsic Pancharatnam-Berry phase that breaks mode degeneracy and elevates sensing sensitivity.
- **Visual Layout**: Side-by-side comparative architectural schematic illustrating field distributions on a standard ring resonator versus a topological Möbius strip resonator.
- **Physical Comparison**:
    - *Standard Ring Resonator*: Possesses an orientable two-sided surface with zero Berry phase (\gamma = 0); supports degenerate clockwise and counter-clockwise integer whispering-gallery modes (f_m \propto m); exhibits low polarization selectivity.
    - *Möbius Strip Resonator*: Possesses a non-orientable single-sided surface enforcing a 180^\circ polarization twist over physical length L; wave must complete 2L to close the loop, accumulating a geometric Berry phase of \pi; lifts mode degeneracy to establish fractional half-integer resonances (f_m \propto m + 1/2) that resonate selectively with single circular polarization states.

Slide 6: Field-Directed Biofabrication & Printhead Hardware

- **Slide Title**: Dynamic Roto-Field Micro-Extrusion Hardware
- **Core Message**: High-precision orientation of synthetic silk and aramid nanocomposites locked via in-situ photopolymerization.
- **Visual Layout**: Exploded mechanical cutaway of the ScanIQ extrusion printhead assembly with component callout tags and an inset SEM microstructure graphic.
- **Hardware Assembly Callouts**:
    - *Micro-Aperture Extrusion Nozzle*: Fluidic channel (50\text{ to }150\text{ }\mu\text{m} inner diameter) delivering UV-curable methacrylated silk fibroin / ANF feedstock.
    - *Dynamic Roto-Field Collar*: Micro-solenoid stator array generating rotating magnetic vector fields (0.1\text{ to }0.5\text{ T}, 1\text{ to }10\text{ Hz}) to program fiber alignment.

    - *Coaxial Photopolymerization Ring*: Multi-wavelength UV/blue LED array (365\text{ nm} / 405\text{ nm}) providing immediate crosslinking within milliseconds of deposition.
    - *Aligned Microstructure Inset*: Microscopic view illustrating continuous orientation of high-modulus aramid nanofiber helices along the print path.

Slide 7: Sensor Kernel: Opposed Near-Field Interrogation

- **Slide Title**: Opposed Resonator Microfluidic Cassette
- **Core Message**: Near-field evanescent wave coupling bypasses liquid water absorption while integrated diamond NV magnetometry captures bio-magnetic transients.
- **Visual Layout**: Vertical architectural cross-section of the biological sensing cassette illustrating the layered physical components and the isolated reference channel.
- **Cassette Architecture Directives**:
    - *Upper Metasurface*: Input waveguide horn coupling polarization-modulated terahertz pulses into the upper Möbius resonator.
    - *Microfluidic Incubation Channel*: Low-loss cyclo-olefin polymer cassette (120\text{ }\mu\text{m} height) holding the embryo in physiological media.
    - *Evanescent Interaction Tail*: Evanescent field decay (E \propto e^{-\kappa z}) confined to the cell boundary (z < 10\text{ }\mu\text{m}), entirely avoiding bulk liquid water attenuation.
    - *Lower Metasurface*: Opposed Möbius strip capturing perturbed chiroptical transmission and Q-factor degradation.
    - *Biological Quantum Node*: NV diamond substrate beneath the trap recording local vector fields.
    - *Synthetic Reference Node*: Spatially isolated NV diamond node (30\text{ mm} lateral distance) recording common-mode Schumann resonances and laboratory EMI for real-time cancellation.

Slide 8: AI Discriminator: The Digital Developmental Phenotype

- **Slide Title**: Trajectory Modeling in Multi-Dimensional Phase Space
- **Core Message**: ScanIQ replaces static threshold grading with continuous temporal trajectory analysis through a multimodal physical phase space.
- **Visual Layout**: Multi-channel temporal trajectory graph comparing a normally developing embryo against an arrested embryo across a 120\text{-hour} culture window.
- **Analytical Insights Displayed**:
    - *Normal Developmental Trajectory*: Smooth, monotonic decrease in 1.49\text{ MHz} real impedance tracking cellular mass expansion; rhythmic, cyclic Q-factor oscillations matching mitotic divisions; stable low-frequency membrane resistance.
    - *Arrested / Non-Viable Trajectory*: Early inflection away from the normative manifold at 36\text{ hours} (premature membrane depolarization and abnormal Q-factor damping), triggering an alert 18\text{ hours} before optical cleavage cessation or blastomere blebbing becomes visible.
    - *Dual-Atlas Architecture*: Independent training paths for Gamete Meiotic Maturation and Embryo Mitotic Cleavage prevent biological confounding.

Slide 9: Staged Empirical Validation Roadmap

- **Slide Title**: Structured Clinical and Metrological De-Risking
- **Core Message**: A five-phase technical validation plan establishing measurement reproducibility, safety, and diagnostic efficacy.
- **Visual Layout**: Five-stage horizontal progression table detailing specific validation objectives, experimental models, and clear pass/fail exit gates.

Stage	Focus Area	Experimental Model & Methods	Target Exit Gate / Validation Criteria
Stage 1	Materials & Resonator Calibration	3D-printed silk/ANF coupon libraries; VNA and THz time-domain spectroscopy.	Flexural strength \ge 500\text{ MPa}; loaded cavity Q > 150 at 1.0\text{ THz}.
Stage 2	Synthetic Phantom Validation	Calibrated dielectric polystyrene microspheres; micro-droplets of graded salinity.	Linear differential frequency shift (\Delta f_r \propto \Delta\epsilon); zero thermal heating.
Stage 3	Non-Human Model Embryology	Blinded bovine and porcine *in vitro* produced embryos in microfluidic traps.	Replicate 1.49\text{ MHz} impedance curve; achieve >85\% stage prediction accuracy.
Stage 4	Multimodal Clinical Correlation	Discarded/donated human embryos under institutional reproductive research oversight.	Fused model demonstrates \text{AUC} \ge 0.92 against blastulation and euploidy status.
Stage 5	Regulated Translation	Multi-center prospective observational trials across accredited IVF clinics.	Regulatory clearance (510(k)/PMA); clinical non-inferiority demonstrated.

Slide 10: Intellectual Property Portfolio Architecture

- **Slide Title**: Multi-Tiered Patent Protection Strategy
- **Core Message**: Broad, defensible intellectual property coverage spanning manufacturing algorithms, hardware design, quantum denoising, and predictive software.
- **Visual Layout**: Two-by-two matrix categorizing core patent claims, target filings, and commercial barriers to entry.
- **IP Protection Pillars**:
    - *Pillar 1 (Additive Manufacturing)*: Claims covering SDF and continuous material field generation, micro-aperture printheads, and dynamic roto-field alignment of synthetic silk/aramid composites.
    - *Pillar 2 (Metamaterial Hardware)*: Claims covering non-orientable topological Möbius strip resonators, opposed near-field evanescent biosensing channels, and microfluidic cassettes.

    - *Pillar 3 (Quantum Sensing Architecture)*: Claims covering differential NV diamond ODMR magnetometry, spatial gradiometer configurations, and real-time Schumann resonance baseline cancellation.
    - *Pillar 4 (Predictive AI Models)*: Claims covering temporal cross-attention transformers, decoupled meiotic/mitotic spectral atlases, and early trajectory deviation scoring.

Slide 11: Market Opportunity and Business Model

- **Slide Title**: Scalable Razor / Razor-Blade Commercial Architecture
- **Core Message**: High-margin recurring consumables and SaaS revenue built upon a global unmet need in reproductive medicine.
- **Visual Layout**: Three-part commercial breakdown highlighting Total Addressable Market (TAM), three-tier revenue streams, and health economic ROI.
- **Commercial Metrics**:
    - *Total Addressable Market*: Global IVF and ART market projected at 25.8\text{B} by 2030 (10.2% CAGR), with over 3.2\text{ million} cycles performed annually worldwide.
    - *Hardware Capital Sale*: ScanIQ Bio-Resonance Incubation Chamber and optical/quantum transceiver unit ($120,000 capital purchase).
    - *High-Margin Consumables*: Single-use, 3D-printed microfluidic cassettes with integrated topological resonators ($350 per patient cycle, 85\% gross margin).
    - *Recurring SaaS Licensing*: ScanIQ Cloud AI developmental trajectory reporting and viability scoring ($150 per analyzed embryo cycle).
    - *Clinical Economic Value*: Increasing first-cycle live-birth pregnancy rates by an absolute 5\text{ to }8\% saves intended parents over $12,000 in avoided repeat cycles.

Slide 12: Partnership Ask and 12-Week Feasibility Sprint

- **Slide Title**: Accelerating Commercial Validation
- **Core Message**: Seeking 3.5\text{M} in Seed financing and academic co-development partnerships to execute a 12-week empirical feasibility sprint.
- **Visual Layout**: Structured 12-week operational Gantt schedule outlining immediate deliverables across materials, hardware, biology, and data science.
- **12-Week Sprint Deliverables**:
    - *Weeks 1–4 (Materials & AM)*: Formulate Sil-MA/ANF resin; print 100 coupon prototypes; verify 500\text{ MPa} flexural strength and topological mode resonance.
    - *Weeks 5–8 (Cassette & Quantum Integration)*: Assemble opposed near-field cassettes; integrate dual NV-diamond nodes; validate Schumann resonance rejection.
    - *Weeks 9–12 (Biological Demonstration)*: Execute 50-embryo bovine validation run; demonstrate real-time 1.49\text{ MHz} tracking and early trajectory divergence detection.
- **Contact Information**: Executive Management Team | ScanIQ LLC | info@scaniq.bio.

9. Provisional Patent Application Draft

Specification

Title of the Invention

ADDITIVE MANUFACTURING OF TOPOLOGICAL CHIRAL METAMATERIAL RESONATORS FOR NEAR-FIELD TERAHERTZ SPECTROSCOPY AND QUANTUM-ASSISTED DEVELOPMENTAL PHENOTYPING

Cross-Reference to Related Applications

This application claims the benefit of priority to U.S. Provisional Patent Application Serial No. 63/xxx,xxx, filed September 8, 2026, the entire contents of which are incorporated herein by reference in their entirety.

Field of the Invention

The present disclosure relates generally to advanced additive micromanufacturing, topological metamaterials, and non-invasive biomedical metrology. More specifically, the disclosure relates to systems and methods for additive manufacturing of functionally graded, chiral silk-nanofiber topological resonators using dynamic field-assisted extrusion, and utilizing said resonators in opposed near-field terahertz polarimetry and nitrogen-vacancy diamond quantum magnetometry to assess biological viability non-destructively.

Background of the Invention

In assisted reproductive technology (ART) and *in vitro* fertilization (IVF), accurately evaluating gamete and preimplantation embryo quality is critical for clinical success. Existing optical time-lapse incubation systems assess developmental potential based entirely on exterior morphology and the timing of cellular cleavage. These visible markers, however, represent late-stage physical manifestations of cellular health; metabolic exhaustion, ionic dysregulation, and cytogenetic damage often occur hours or days before optical changes appear. Conversely, molecular and genetic testing, such as preimplantation genetic testing for aneuploidies (PGT-A), requires invasive trophectoderm micro-biopsy and mandatory cryopreservation, which introduces micro-mechanical trauma, biological stress, and significant economic cost.
While electrical impedance spectroscopy (EIS) has shown promise in detecting changes in embryonic cell mass and physiological integrity, traditional planar electrode configurations suffer from severe electrode polarization impedance, capacitive shunting, and lack of spatial resolution. Furthermore, although high-frequency terahertz radiation can directly probe macromolecular vibrations (chiral phonons), the extremely high dielectric absorption of liquid water in physiological culture media has prevented the continuous, non-destructive optical interrogation of living specimens. There is a critical unmet need for a non-invasive, continuous diagnostic platform capable of monitoring intracellular integrity without physical trauma or water-induced signal loss.

Summary of the Invention

The present disclosure provides a bio-resonance metrology platform that integrates continuous Signed Distance Field (SDF) additive manufacturing, field-aligned biomimetic nanocomposites, topological Möbius strip resonators, evanescent near-field terahertz polarimetry, and differential nitrogen-vacancy (NV) diamond quantum magnetometry.

In one embodiment, a field-assisted additive manufacturing system utilizes a micro-aperture extrusion printhead surrounded by a dynamically rotating electromagnetic field ("roto-field") to align anisotropic aramid nanofibers and silk fibroin within a photopolymerizable resin, generating a topological Möbius resonator. The non-orientable, single-sided topology of the Möbius strip introduces an intrinsic Pancharatnam-Berry phase of \pi that breaks clockwise/counter-clockwise mode degeneracy, establishing fractional half-integer resonances exhibiting high quality factors and circular chiroptical selectivity.

In another embodiment, two opposed topological resonators are positioned above and below a microfluidic incubation channel, interrogating an embryo via evanescent near-field coupling. This evanescent configuration confines the electromagnetic field to the cellular boundary, circumventing bulk liquid water absorption. Simultaneously, dual nitrogen-vacancy diamond quantum magnetometers execute differential spatial gradiometry against environmental baselines, including Earth-ionosphere Schumann resonances (7.83\text{ Hz}), isolating faint biological magnetic signatures. A temporal cross-attention artificial intelligence discriminator processes the resulting multi-modal data to classify developmental stages and predict embryo viability prior to the physical manifestation of morphological defects.

Brief Description of the Drawings

- **Figure 1** is a schematic representation of the field-assisted additive manufacturing apparatus, depicting the micro-aperture printhead, dynamic electromagnetic roto-field stator coils, and coaxial photopolymerization assembly.
- **Figure 2** is a topological diagram of an additive-manufactured Möbius strip resonator, illustrating the non-orientable surface, internal nanofiber director orientation, and the geometric Pancharatnam-Berry phase accumulation.
- **Figure 3** is a cross-sectional architectural view of the opposed-resonator microfluidic sensing cassette, showing the evanescent interaction zone and integrated dual NV-diamond quantum magnetometry substrates.
- **Figure 4** is a block diagram of the terahertz pulse-modulation and polarimetric electro-optic sampling detection chain.
- **Figure 5** is a schematic of the differential quantum gradiometer and adaptive Schumann resonance baseline noise cancellation system.
- **Figure 6** is a block diagram of the Multi-Modal Cross-Attention Temporal Transformer (MCAT-Net) artificial intelligence architecture.
- **Figure 7** is a graphical plot illustrating differential real impedance trajectories at 1.49\text{ MHz} and terahertz Q-factor shifts across normal versus arrested mammalian developmental courses.

Detailed Description of Preferred Embodiments

Referring to Figure 1, the fabrication apparatus comprises an implicit computational geometry compiler executing on a digital processor. The compiler maintains a signed distance function

f(\vec{x}) defining the geometric boundary of a Möbius strip, unified with a continuous material tensor field \mathbf{M}(\vec{x}) dictating internal fiber alignment vectors \vec{v}_{\text{fiber}}(\vec{x}) and local structural chirality \chi(\[span_151](start_span)[span_151](end_span)vec{x}). The feedstock comprises methacrylated silk fibroin reinforced with functionalized aramid nanofibers (achieving flexural strengths \ge 500\text{ MPa} when cured). The feedstock is extruded through a micro-aperture nozzle (100\text{ }\mu\text{m} inner diameter). Concentric micro-solenoid coils generate a dynamically rotating magnetic field (0.1\text{ to }0.5\text{ T}, 1\text{ to }10\text{ Hz}) transverse to the extrusion axis, exerting alignment torques on the suspended fibers. Coaxial UV LEDs (365\text[span_9](start_span)[span_9](end_span){ nm}) initiate in-situ photopolymerization within 50\text{ milliseconds} of deposition, locking the programmed orientation into the solid matrix.

Referring to Figure 2 and Figure 3, the manufactured topological resonator comprises a Möbius strip ribbon of circumference L, width w = 0.5\text{ mm}, and thickness t = 0.1\text{ mm}, characterized by a 180^\circ spatial half-twist. Two identical resonators are mounted in mirror opposition across a microfluidic incubation cassette with a channel height of 120\text{ }\mu\text{m}. A mammalian embryo rests in a localized trapping depression between the resonators. The near-field evanescent decay tails (E \propto e^{-\kappa z}) penetrate the outer cellular boundary while avoiding bulk liquid water attenuation.

Referring to Figure 4, an ultrafast fiber laser delivers optical pulses to a photoconductive terahertz emitter, generating broadband pulses spanning 0.2\text{ to }2.5\text{ THz}. High-speed polarization modulators alternate the pulse train between left- and right-circular polarization states. The transmitted signals are collected by an electro-optic sampling detector, which records time-resolved Stokes vectors [S_0, S_1[span_97](start_span)[span_97](end_span), S_2, S_3] to extract differential circular dichroism (\Delta S_3) and quality factor degradation (\Delta Q).

Referring to Figure 5, quantum bio-magnetic sensing is executed by two optical nodes comprising diamond substrates with ensembles of nitrogen-vacancy (\text{NV}^-) centers. A 532 nm laser optically pumps the ensembles, and microwave striplines drive optically detected magnetic resonance (ODMR) across the 2.87\text{ GHz} zero-field splitting. The sensing node is positioned within 50\text{ }\mu\text[span_111](start_span)[span_111](end_span)[span_117](start_span)[span_117](end_span){m} of the embryo trap, while the reference node is positioned 30\text{ mm} away. An adaptive FIR Wiener filter cancels the common-mode 7.83\text{ Hz} Earth-ionosphere Schumann resonance and laboratory power-line noise, isolating the embryonic magnetic transients.

Referring to Figure 6 and Figure 7, the digital phenotyping engine streams the electrical impedance spectra Z(f, t), resonant frequency shifts \Delta f_r(t), quality factors Q(t), terahertz circular dichroism, vector magnetic fields \vec{B}(t), and optical morphokinetics into a temporal cross-attention transformer. The model tracks developmental trajectories against an embryonic mitotic cleavage atlas. When an embryo undergoes metabolic arrest or fragmentation, the differential real impedance at 1.49\text{ MHz} fails to follow the normative cellular mass expansion curve, and the terahertz Q-factor exhibits abnormal damping, generating a predictive failure score up to 18\text{ hours} before optical fragmentation manifests.

Patent Claims

What is claimed is:

1. A resonant biofabrication and developmental sensing system, comprising: an additive manufacturing apparatus comprising an extrusion aperture, an electromagnetic orientation assembly configured to apply a dynamically rotating vector field within an extrusion print zone, and an irradiation source configured to initiate in-situ photopolymerization of an extrudable composite feedstock containing anisotropic microstructures to fabricate a solid topological resonator; a sensing fixture comprising at least one manufactured topological resonator positioned adjacent to a biological incubation chamber; an electromagnetic source operably coupled to pump modulated high-frequency signals through the topological resonator to establish an evanescent near-field within the biological incubation chamber; and a detection subsystem configured to measure perturbations in a quality factor (Q-factor), an eigenmode resonance frequency, and a polarization state of the topological resonator induced by a biological specimen positioned within the biological incubation chamber.
2. The system of claim 1, wherein the topological resonator comprises a Möbius strip geometry having a single-sided non-orientable surface and an accumulated geometric Pancharatnam-Berry phase of \pi, supporting fractional half-integer electromagnetic modes.
3. The system of claim 1, wherein the extrudable composite feedstock comprises regenerated silk fibroin, aramid nanofibers (ANFs), and a photoinitiator, wherein the cured topological resonator exhibits a flexural strength of at least 450\text{ MPa}.
4. The system of claim 1, wherein the electromagnetic orientation assembly comprises a multi-pole stator array that generates a rotating magnetic vector field at a frequency between 0.1\text{ Hz} and 100\text{ Hz}, orienting the anisotropic microstructures along a three-dimensional vector field defined by an implicit signed distance function.
5. The system of claim 1, wherein the electromagnetic source comprises a pulsed terahertz emitter operating within a frequency range of 0.1\text{ THz} to 3.0\text{ THz}, and wherein the detection subsystem comprises a polarimetric electro-optic sampling detector configured to measure time-resolved Stokes parameters [S_0, S_1, S_2, S_3].
6. The system of claim 1, further comprising a quantum magnetometry subsystem, the subsystem comprising: a primary diamond substrate comprising an ensemble of nitrogen-vacancy (NV) color centers positioned adjacent to the biological incubation chamber; a secondary reference diamond substrate comprising an ensemble of NV color centers spaced apart from the biological incubation chamber; and a differential processing unit configured to perform optically detected magnetic resonance (ODMR) vector magnetometry and execute adaptive common-mode cancellation of environmental baseline electromagnetic fields.
7. The system of claim 6, wherein the environmental baseline electromagnetic fields canceled by the differential processing unit comprise Earth-ionosphere Schumann resonances having a fundamental frequency of approximately 7.83\text{ Hz}.
8. The system of claim 1, further comprising an artificial intelligence discriminator stored in a non-transitory memory and executed by a processor, configured to: ingest time-series streams of the measured Q-factor, resonance frequency, polarization state, wideband electrical impedance spectra, and optical time-lapse morphokinetics from the biological specimen; and map the time-series streams against a calibrated developmental trajectory to output a predictive score of biological viability.
9. The system of claim 8, wherein the wideband electrical impedance spectra include differential real impedance measurements sampled across a frequency band encompassing 1.49\text{ MHz}.

10. A method for non-invasively assessing biological developmental viability, comprising: positioning an embryonic specimen within a microfluidic sensing well positioned between an opposed pair of 3D-printed topological Möbius resonators; coupling polarization-modulated terahertz pulses into the opposed topological Möbius resonators to generate an evanescent electromagnetic field that interacts with the embryonic specimen; measuring shifts in a complex resonance frequency (\Delta f_r), a quality factor (\Delta Q), and a terahertz circular dichroism (\Delta S_3) parameter across a developmental monitoring period; measuring localized vector magnetic field transients using a differential nitrogen-vacancy diamond quantum magnetometer, canceling ambient Schumann resonance baselines; and processing the measured resonance shifts, quality factor, circular dichroism, vector magnetic field transients, and synchronized optical time-lapse frames through a temporal transformer model to calculate a developmental trajectory deviation metric.
11. The method of claim 10, wherein the opposed pair of topological Möbius resonators are additive-manufactured from a synthetic silk fibroin and aramid nanofiber composite whose internal molecular director orientation is steered via a rotating electromagnetic field during micro-extrusion.
12. The method of claim 10, wherein the temporal transformer model maintains separate, decoupled latent training representations for gamete meiotic maturation and embryonic mitotic cleavage.
13. A non-transitory computer-readable storage medium storing instructions that, when executed by one or more processors, cause the one or more processors to: compile an implicit signed distance field (SDF) and a spatial material orientation field representing a topological Möbius resonator; translate the SDF and spatial material orientation field into synchronized machine toolpaths for an additive manufacturing apparatus, controlling vertical printhead translation, in-situ photopolymerization dosage, and dynamic multi-axis coil currents in an electromagnetic orientation head; ingest multi-modal sensory data comprising terahertz circular dichroism, resonator Q-factor, wideband electrical impedance, and nitrogen-vacancy quantum magnetometry signals from a biological specimen positioned adjacent to the topological Möbius resonator; adaptively filter common-mode environmental noise from the quantum magnetometry signals utilizing an Earth-ionosphere Schumann resonance baseline model; and compute an embryonic developmental stage classification and a viability prediction score utilizing a multi-modal cross-attention neural network.

10. Strategic Conclusions and Commercialization Outlook

The synthesis of field-directed additive micromanufacturing, high-strength biomimetic nanocomposites, non-orientable topological metamaterials, and quantum-assisted polarimetric sensing establishes a viable pathway toward non-invasive, high-content biological metrology. By moving past the physical limitations of bulk terahertz transmission and subjective optical inspection, the platform captures the continuous electromechanical phenotype of development without inducing mechanical or radiation stress.

The immediate translational trajectory focuses on executing a structured, five-stage validation roadmap:

- The materials and fabrication phase must demonstrate consistent extrusion and

crosslinking of aramid/silk composites with flexural strengths exceeding 500\text{ MPa} and loaded terahertz cavity Q-factors above 150.
- The microfluidic sensing cassette must undergo rigorous bench testing with calibrated synthetic phantoms to confirm that the opposed evanescent near-field architecture eliminates bulk liquid water attenuation while maintaining linear sensitivity to dielectric perturbations.
- Preclinical validation in established bovine and porcine model systems must replicate the 1.49\text{ MHz} differential impedance developmental inflection and establish multi-modal baseline atlases under blinded conditions.
- Algorithmic training must enforce strict separation between gamete meiotic atlases and embryonic mitotic cleavage trajectories, verifying that the temporal transformer detects developmental arrest hours before optical manifestations appear.
- Finally, prospective clinical observational trials conducted under reproductive research oversight will benchmark the platform's predictive accuracy (\text{AUC} \ge 0.92) against established clinical endpoints.

Through this closed-loop engineering framework—linking implicit geometry, field-directed material synthesis, topological electromagnetics, and deep temporal intelligence—the ScanIQ platform defines an objective, quantitative standard for non-destructive developmental analysis in reproductive medicine.

Works cited

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